Cpts 317. 11/9

Summary:

① This is an alg to solve emptiness of PDA.

Given: PDA M

Question: $L(M) = \emptyset$ ?

② c.f. languages are closed under intersection with reg languages.

On Final Exam.

Show: Let M be a FA. There is an alg to decide whether there is a word w s.t.

$$w \in L(M) \text{ and } \#_a(w) = \#_b(w).$$

Bad proof. (0/100). My alg is:

Input FA M, word w

NOT given!

Check: Run M on w and make sure M says yes on w, and I count $\#_a(w)$, $\#_b(w)$ and make sure $\#_a(w) = \#_b(w)$.

not part of input.

Good proof. We construct

$$\mathcal{L} = L(M) \cap \{w : \#_a(w) = \#_b(w)\}.$$

Observe that, since $L(M)$ is regular and

$\{w : \#_a(w) = \#_b(w)\}$ is c.f., and hence

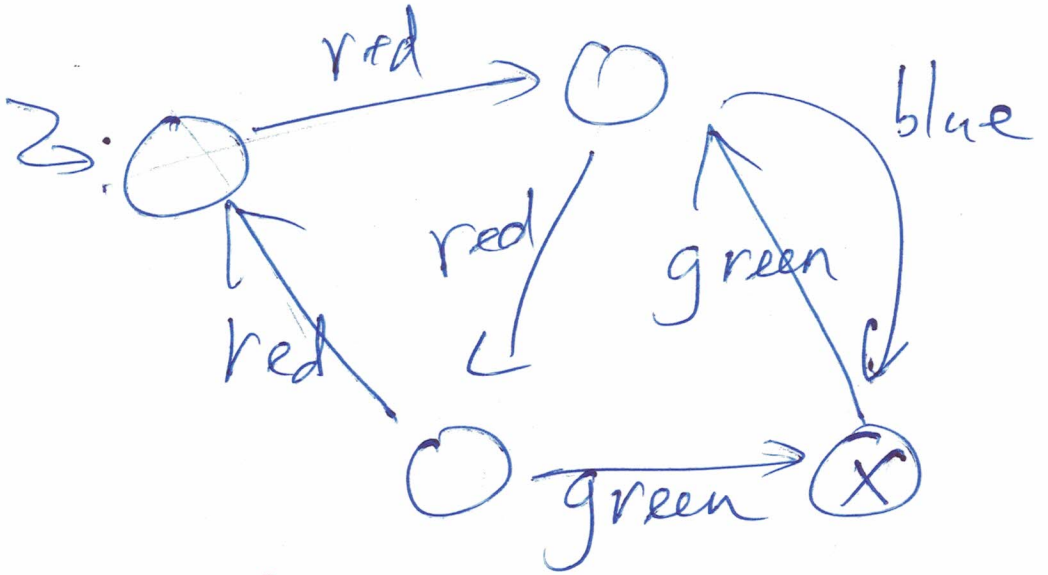
$\mathcal{L}$ is c.f. (and accepted by a PDA). The result follows since we have alg to solve "$\mathcal{L} = \emptyset$?".

On final Exam

Let $G$ be a graph where each edge has a color and multiple edges can share the same color. There is a designated start node and a designated end node in $G$. Let $w$ be a walk from start to end. We use $\#_{red}(w)$ to denote the number of red edges on $w$. Show that: there is an alg to decide whether there is a walk $w$ from start to end in $G$ sat:

$$\#_{red}(w) = \#_{green}(w).$$

Small Example:




// walk can
// contain cycles
// and hence its
// length is unbounded

Bad alg. (0/100)

Input: graph $G$, a walk $W$ → not part of input.

Check: $\#_{red} == \#_{green}$ on $W$.

if yes, ret yes,

ret No.

Bad alg. (1/100). my alg:

input: G

Enumerate all walks w from start to end and check one by one:

if $\#_{red}(w) == \#_{green}(w)$

ret yes

ret NO.

how many? could be infinite.

Good proof. I define (color sequences on)

$\mathcal{L}_1$ = the set of all walks from start to end in $G$;

$\mathcal{L}_2$ = the set of all color sequences with $\#_{red} = \#_{green}$.

Notice that $\mathcal{L}_1$ is regular and $\mathcal{L}_2$ is c.f. Then $\mathcal{L} = \mathcal{L}_1 \cap \mathcal{L}_2$ is c.f. The result follows since we have alg to solve "$\mathcal{L} = \phi$?".

On final Exam.

Try: previous problem with added constraint:

$$\#_{red}(\omega) = \#_{green} \quad \text{and}$$

$$\#_{red} \bmod 17 = 2.$$

Let $P$ be a C-program where we have only one integer var $x$. We can only operate $x$ using the following instructions:

$x$++;

$x$--;

$x$==0?,

along with goto statement.

Example program:

1. x++;
2. if x==0 then goto 3 else goto 1
3. x--;
4. goto 2
5. HALT.

P halts if, when P runs from $x=0$, it will reach the HALT-statement. Shows that there is an alg to decide whether P halts.

Bad proof. (0/100). My alg is:

Input P —— program

Run P with initial val $x=0$;

Watch it! if P halts, get yes,

P could run forever. get No.

Good proof. Let $\underline{P}$ be the given program. We construct a PDA M to simulate the $\underline{P}$ as follows:

M never reads input. When $\underline{P}$ does

$x$++ , M pushes $\delta$ into the stack;

$x$-- , M pops $\delta$ from the stack;

$x$==0? , M checks if the stack is empty;

Note that program counter in $\underline{P}$ is stored in finite memory of M. Also notice that, WLOG, $x$ is nonnegative; o.w, sign of $x$ can be stored also in M's finite memory. When $\underline{P}$ reaches HALT, M says yes.

Clearly, $L(M) = \emptyset$ iff $P$ doesn't halt. The result follows since we have alg to decide "$L(M) = \emptyset$?".

On final Exam.

Try the previous problem with additional instructions:

$x := x + 17;$

$x := x - 5;$

$x == 3?;$

1. Reg. & FA

2. C.f. & PDA

3. Turing machines. ⇐ From Next lecture